

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Architected the deployment-gating layer for production AI releases: automated compliance, latency, and drift checks that block regressions before they reach customers, catching 15% of releases that failed validation criteria.
- Led design and delivery of a multi-agent orchestration system (OpenAI Agents SDK, MCP tool-calling, RAG over Pinecone), owning it end-to-end from architecture through production monitoring at 99%+ uptime and sub-200ms latency.
- Set the infrastructure standard for the team's AI workloads: serverless architecture (AWS Lambda, S3, DynamoDB, API Gateway, SageMaker) supporting high-concurrency inference at production scale.

QureAI (AI Startup)

Jan 2024 – May 2025

ML Infrastructure Engineer

- Owned an AI system from evaluation through fix: diagnosed a retrieval-quality gap from production signals, redesigned the RAG pipeline, and lifted recall 61% to 89% while cutting the underlying complaint rate 22% to 6%.
- Led migration of model serving from synchronous to async vLLM with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests through architecture and deployment changes.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices in a distributed, fault-tolerant system for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Mentored engineers and set technical direction for failure-handling standards across 25+ services, translating architecture decisions for cross-functional stakeholders and reducing production escalations 60% in one quarter.

Projects

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with Pinecone vector retrieval and hybrid re-ranking decomposes multi-table questions into dependency-ordered SQL subqueries; deployable as MCP-compatible server for enterprise data queries.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Java, TypeScript, SQL

AI Architecture: Agent Orchestration (LangGraph, OpenAI Agents SDK, MCP), RAG, Vector DBs (Pinecone, FAISS), Prompt Engineering, Model Evaluation & Drift Detection, vLLM

Backend & APIs: FastAPI, REST APIs, Microservices, PostgreSQL

Cloud & Ops: AWS (Lambda, EC2, S3, RDS, SQS, SageMaker), GCP (Cloud Run), Docker, Kubernetes, CI/CD, Production Monitoring

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer